JINGZHANG RISING

A Century of Self-Reliance, Rising in the Shape of 人 (ren, "people")

A3 Design Booklet

Centennial Jing-Zhang AI Innovation Belt Urban Design Open Call

Submission package: cf3901646 | Author: Charlie JIANG | Licence: COMMUNITY-DISPLAY-ONLY

- Core judgment: every spatial metric is recomputed from this package's geometry/ layers in EPSG:4548.
- Boundary status: provisional rough substitute extent, conceptual recommendations only, not statutory control.
- AI public governance: a human fallback channel and a public report card are both mandatory.

The design scope and the three key areas use the organiser's provisional rough substitute boundary: concept recommendations only, never a statutory redline, an approval basis or a precise-area basis. Once official polygons are published the geometry, metrics, drawings and HTML must all be recomputed.

1. Strategic Positioning: China's AI Origin Corridor

In 1909 the Beijing-Zhangjiakou Railway was completed as China's first trunk railway surveyed, designed and constructed independently. Zhan Tianyou resolved the Guangou climb with the 人-shaped switchback at Qinglongqiao, writing the Chinese character for " people" into the spatial memory of engineering self-reliance.

(Qinglongqiao lies outside this design area; the switchback is cited as a cultural motif only, with no geographic conflation.)

More than a century later this same corridor hosts the highest density of top-tier AI researchers, leading models and computing resources in China. The proposal lifts that historical spirit: from "independently building railways" to "independently creating AI intelligence", and from a "human-scale switchback" to "human-centric upward innovation".

2. Core Structure: One Spine, Three Terraces, Two Wings, Seven Stitches

- One main spine: 9.0 km continuous Jing-Zhang Heritage Park greenway with a dedicated commuter bicycle expressway.
- Three terraces: Origin Terrace (Zhongzhiyuan, AI acceleration), Origin Works (incubation and open source), Bell Hub (Dazhongsi, green compute cluster).
- Two wings: Zhongguancun Tech Service Wing (west, fundamental research) and Xiaoyuehe Scenario Wing (east, full-domain testing).
- Seven stitches: 7 east-west pedestrian stitch links healing railway severance and reconnecting university campuses.

3. Governance Covenant

- No AI public service goes live before algorithm filing, a scenario whitelist permit and a human-fallback drill.
- The Report Card Kiosk publishes the six F1-F6 states physically and stays readable with the power off.

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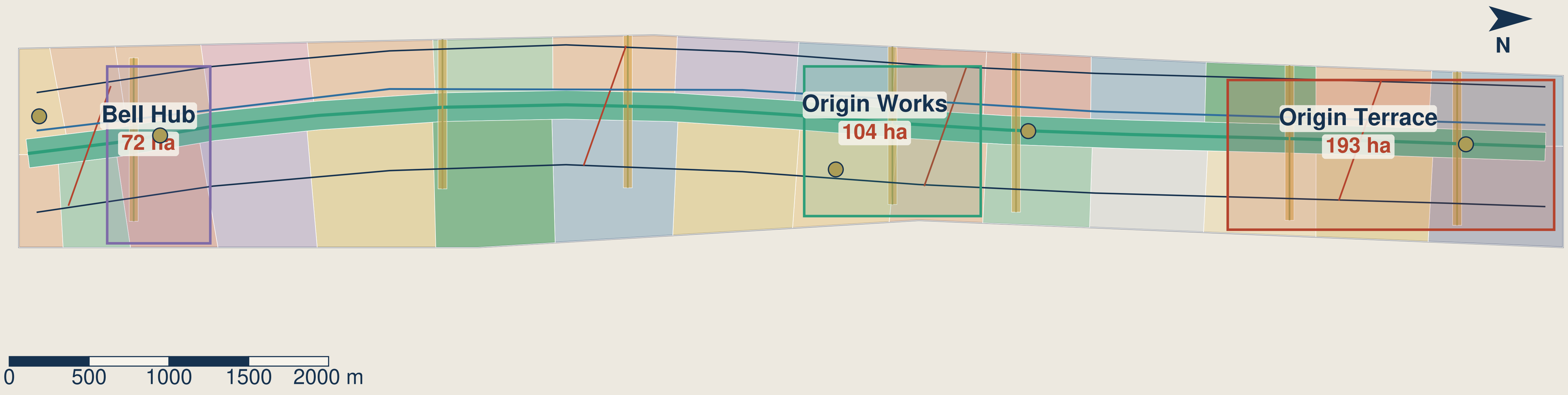
Overall Design Area - Site Overview Plan

Boundary, land use, green space and the three Key Areas are projected from this package's geometry/ layers; the boundary is the organiser's provisional rough extent, not a red line.



Overall design area (provisional) | Key Area | Heritage park spine greenway | Principal greenway (slow-mobility spine)

Commuter cycleway | East-west stitch pedestrian link | Secondary street | Smart-station connector | Pilgrimage landmark

Design area: approx. 11.41 km² (provisional design model; recomputed when the official boundary is issued)

site-overview

Fig. 01 Structure and landmarks

The overall design scope spans approx. 11.41 km² along the historical railway corridor (provisional design model).

Key elements

- 4 pilgrimage landmarks:
  - Renzi Terrace (1909, Qinghe)
  - Gaokao Gate (1949, Tsinghuayuan)
  - Clock of Compute (1980, Dazhongsi)
  - Origin Stele (2026, Origin Works)
- Spatial spine: the Jing-Zhang Heritage Park greenway.
- Regional linkage: Huairou Science City, Future Science City and Yizhuang BDA.

The area figure has a single source, metrics.json site_area_sqm; every carrier states approx. 11.41 km².

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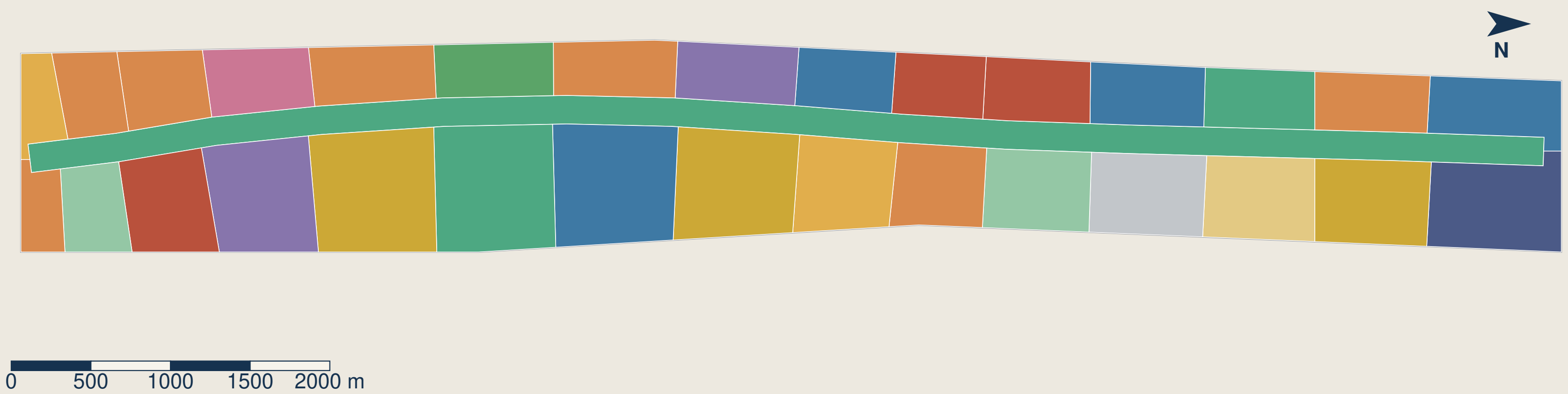
Land-Use Structure Plan (block level)

31 parcels coloured by land-use code; legend and area statistics recomputed from land_use.geojson.



Park and green space · Commercial and business services · Urban residential · Research and development · Cultural · Education · Buffer green · Civic plaza · Public administration and services · Reserved land · Community service facilities · Health care · Sports

Land-use area statistics (recomputed from land_use.geojson)

Code	Land-use category	Area (ha)	Share
1401	Park and green space	**257.1**	22.5%
05	Commercial and business services	**169.7**	14.9%
0701	Urban residential	**147.0**	12.9%
0802	Research and development	**138.7**	12.2%
0803	Cultural	**81.8**	7.2%
0804	Education	**73.1**	6.4%
1402	Buffer green	**55.1**	4.8%
1403	Civic plaza	**49.4**	4.3%
08	Public administration and services	**45.7**	4.0%
16	Reserved land	**36.5**	3.2%
0702	Community service facilities	**35.9**	3.1%
0806	Health care	**26.4**	2.3%
0805	Sports	**25.0**	2.2%

Total 1141.3 ha (31 parcels)

land-use-structure

Fig. 02 Land-use partition

Full spatial partitioning into smart-station and innovation anchor catchments; 31 parcels total 1,141.3 ha (approx. 11.41 km²).

Planning principles

- 4 smart-station TODs: a 500 m pedestrian catchment covering the major employment centres.
- 14 functional anchors: full coverage with no gaps and no overlapping polygons.
- Mixed use: ground-floor public interfaces, shared maker spaces and open research courtyards.

Statutory FAR, height cap and total GFA stay 'unknown' until the official control plan is released; the figures shown are concept ranges only.

The design scope and the three key areas use the organiser's provisional rough substitute boundary: concept recommendations only, never a statutory redline, an approval basis or a precise-area basis. Once official polygons are published the geometry, metrics, drawings and HTML must all be recomputed.

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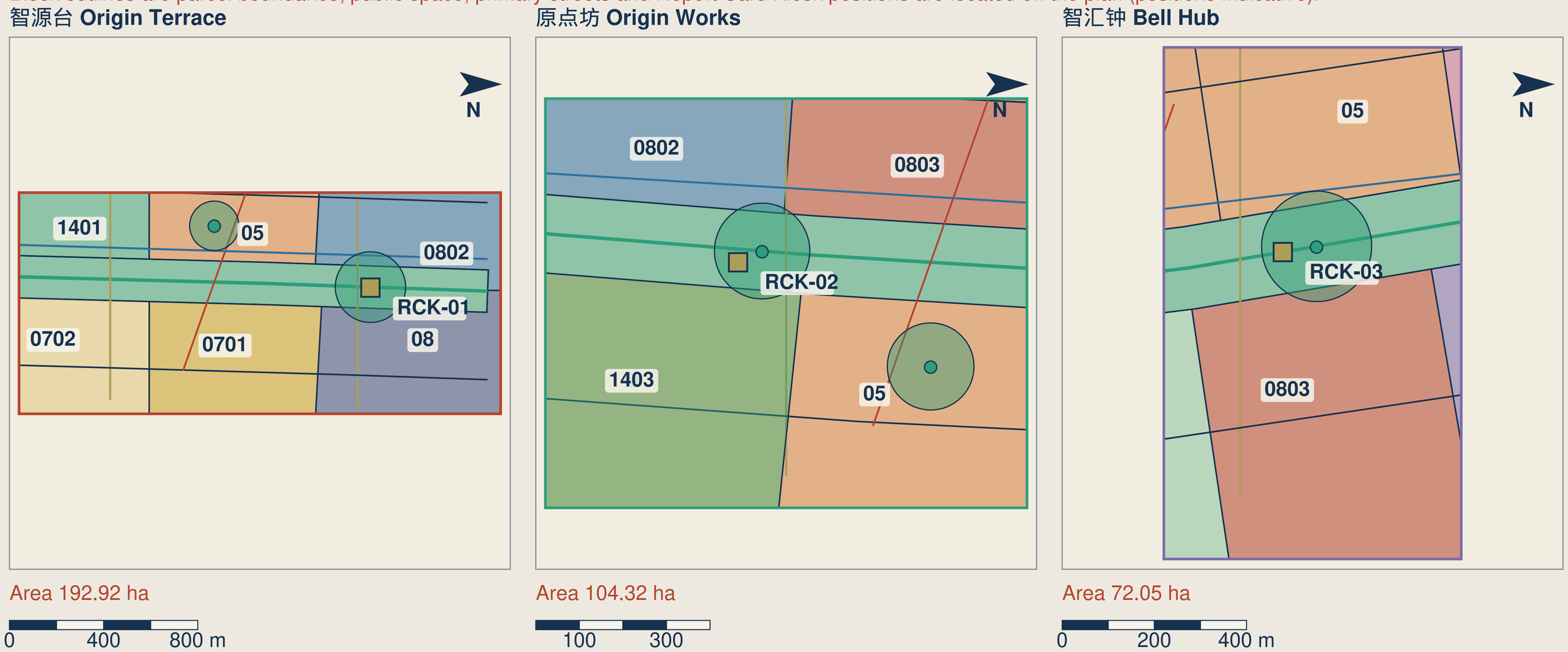
Enlarged Plans of the Three Key Areas

Block outlines are parcel boundaries; public space, primary streets and Report Card Kiosk positions are located on the plan (positions indicative).



Block outline (parcel) | Public space node | Primary street | Report Card Kiosk

Block outlines are labelled with their land-use code; kiosk positions follow the deployment table in proposal.md.

key-areas

Fig. 03 Key areas and blocks

Block-scale design for the three key areas, totalling 369.3 ha.

Three terraces in depth

- Origin Terrace / Zhongzhiyuan (193 ha): ZZ-A..E, FAR 1.6–3.5, height 24–60 m, coverage ≤ 35%.
- Origin Works (104 ha): YD-A..E, FAR 1.2–2.8, height 18–45 m, coverage ≤ 35%.
- Bell Hub / Dazhongsi (72 ha): DZ-A..E, FAR 1.8–3.6, height 24–60 m, coverage ≤ 35%.
- Kiosk deployment: RCK-01, RCK-02 and RCK-03 at the central pedestrian plazas.

These intensities and heights are concept ranges; a professional team must calibrate them once the control plan, tenure and building survey data are available.

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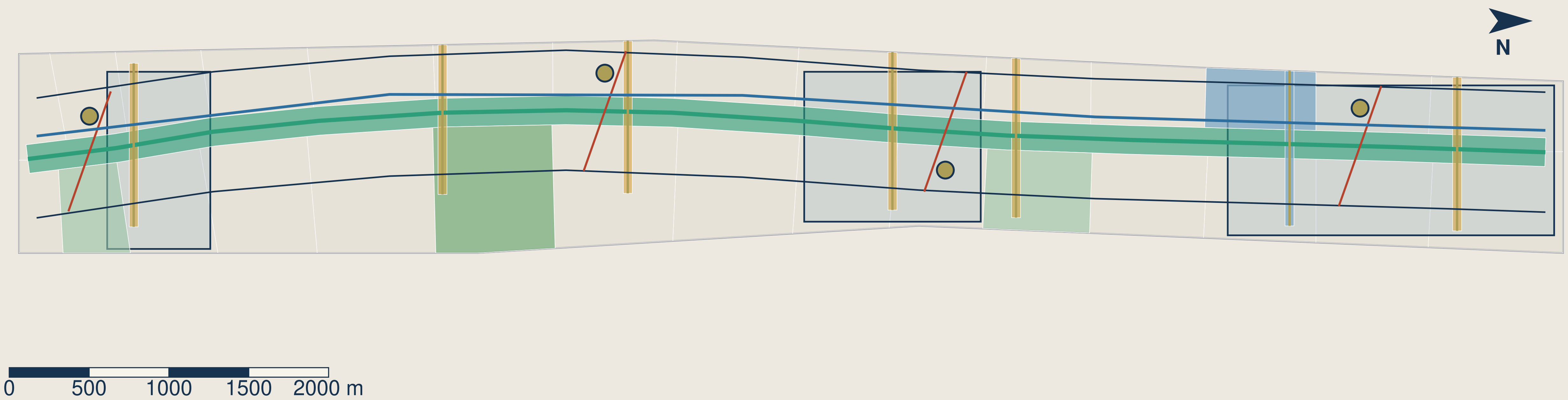
Slow Mobility, Blue-Green and Water Overlay Plan

Cycling network, pedestrian priority, green corridors and waterfront are overlaid on one base plan; each layer is listed in the legend below.



- Principal greenway (slow-mobility spine)
- Commuter cycleway
- East-west stitch pedestrian link
- Smart-station connector
- Secondary street
- Station forecourt public space
- Heritage park spine greenway
- Park and green space
- Buffer green
- Stitch green link
- Waterfront edge (Qinghe; no separate water layer in package)
- Pedestrian priority zone (within Key Areas)

Slow-mobility network 25,932 m (greenway + stitch links + cycleway, recomputed from roads.geojson)

mobility-bluegreen

Fig. 04 Mobility and ecology

A comprehensive pedestrian, bicycle and ecological park corridor network.

Key systems

- 9.0 km main greenway spine with a 4.5 m paved multi-use trail.
- 7 east-west stitch links crossing the former rail barriers.
- Street sections: pedestrian 3–5 m, cycle 2.5–3 m, plus green verges.
- Smart-station curbsides: time-of-day dynamic allocation with EV charging ≥ 40%.

The slow-mobility network totals 25,932 m, recomputed from roads.geojson in EPSG:4548.

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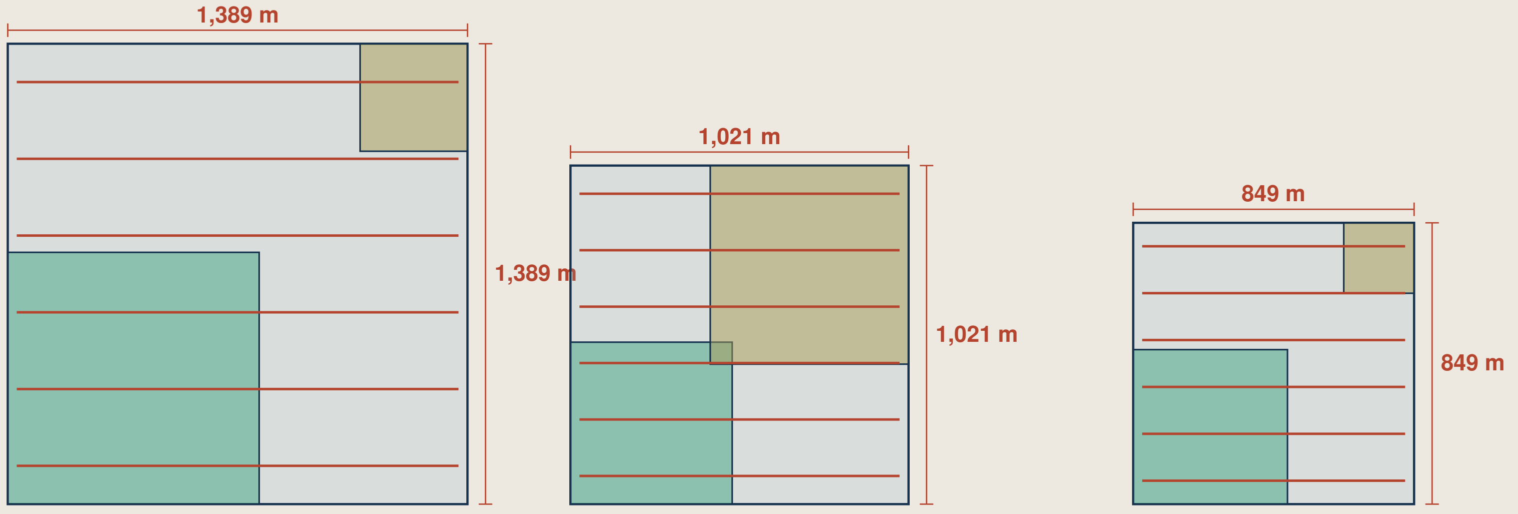
Metrics evidence: equal-area square diagrams of the three key areas

Each square is drawn with the side length of the key area's actual area; areas, lengths and ratios are recomputed from the geometry/ layers



Origin Terrace (Zhongzhiyuan)

Area	192.92 ha
Equivalent side	1,389 m
Slow-mobility length	5,791 m
Slow-mobility density	30.0 m/ha
Green ratio	29.9%
Public space share	5.5%
Building coverage	9.2%
Commuter cycle lane	2,000 m
Station feeder link	797 m

Origin Works (AI Origin Community)

Area	104.32 ha
Equivalent side	1,021 m
Slow-mobility length	3,092 m
Slow-mobility density	29.6 m/ha
Green ratio	22.9%
Public space share	34.4%
Building coverage	8.3%
Commuter cycle lane	1,112 m
Station feeder link	797 m

Bell Hub (Dazhongsi)

Area	72.05 ha
Equivalent side	849 m
Slow-mobility length	2,283 m
Slow-mobility density	31.7 m/ha
Green ratio	30.2%
Public space share	6.3%
Building coverage	10.5%
Commuter cycle lane	653 m
Station feeder link	66 m

Area Green ratio Public space share — Slow-mobility length

Basis: slow mobility = greenway + walking links + cycle lanes; ratios are layer-to-key-area intersections. Boundaries are provisional rough extents.

metrics-evidence

Fig. 05 Metrics and governance

A structured evidence chain recomputing 33 core metrics in EPSG:4548.

Metric invariants

- 30 known metrics: statement value = automated recomputation value.
- 3 statutory controls: FAR, building height and total GFA kept strictly 'unknown'.
- Kiosk typology: footprint 3.96 m² (1,800 × 2,200 × 2,800 mm), clear zone ≥ 2.0 m, wheelchair turning radius ≥ 1.5 m.
- Verification: 100% PASS on all deterministic gates.

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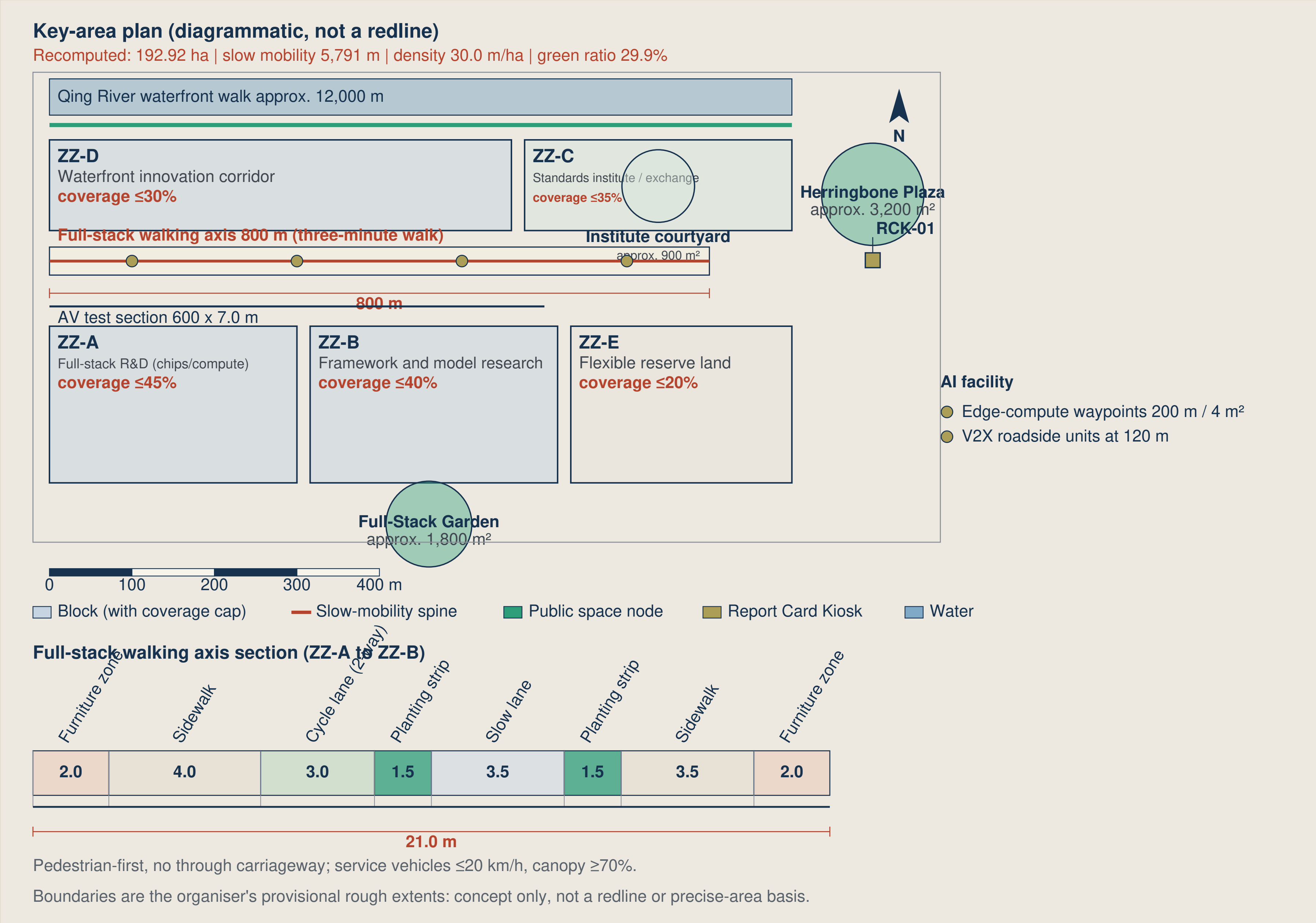

key-areas-zhongzhiyuan

Fig. 06 Origin Terrace, 193 ha

Recomputed: 192.92 ha | slow mobility 5,791 m | density 30.0 m/ha | green ratio 29.9%

- Blocks ZZ-A..ZZ-E: full-stack R&D (chips and compute), framework and model research, standards institute and international exchange, waterfront innovation corridor, flexible reserve land.
- Full-stack walking axis 800 m (a three-minute walk), 21.0 m standard section.
- Public anchors: Herringbone Plaza approx. 3,200 m², Full-Stack Garden approx. 1,800 m², RCK-01 kiosk.
- AV test section 600 × 7.0 m; edge-compute waypoints every 200 m / 4 m².

The Qing River waterfront walk of approx. 12,000 m runs parallel to the main greenway spine.

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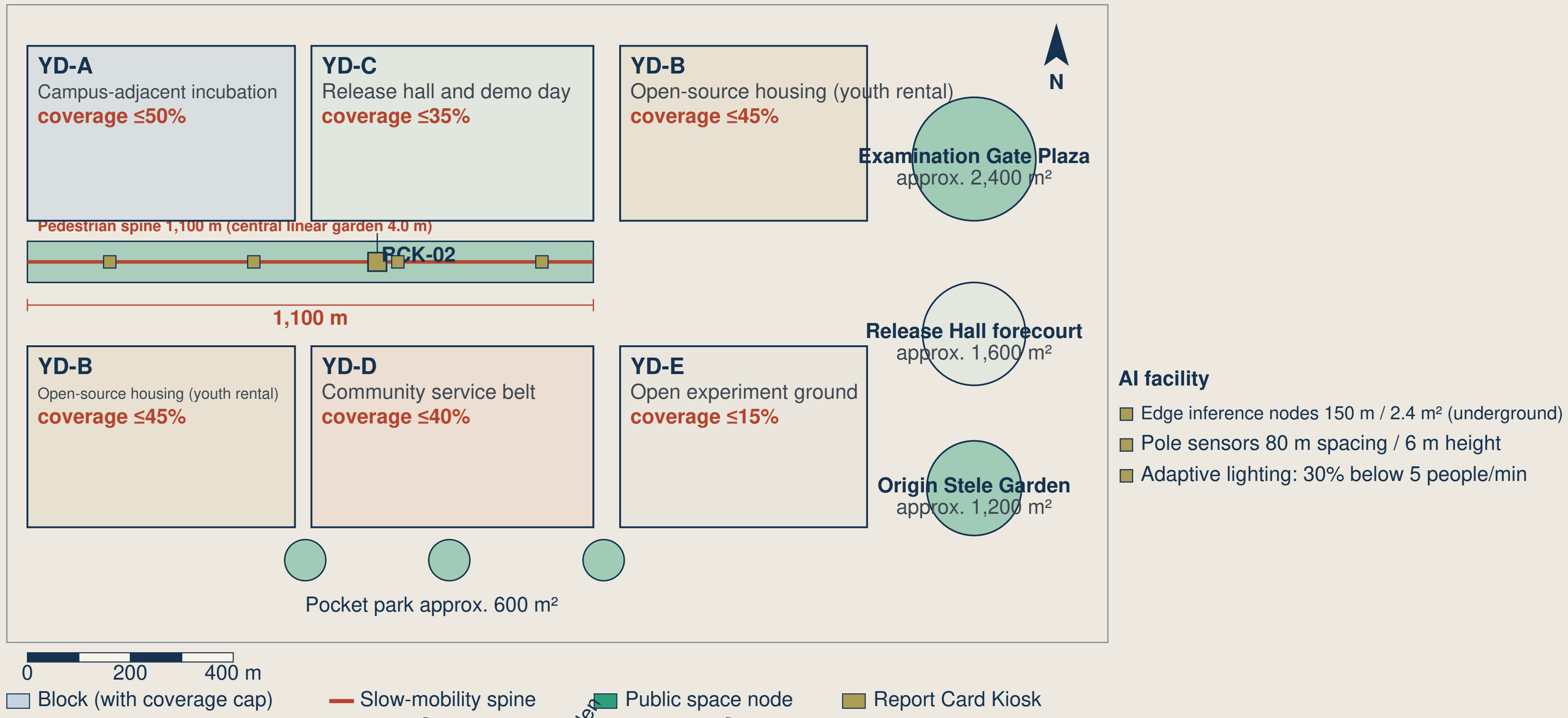
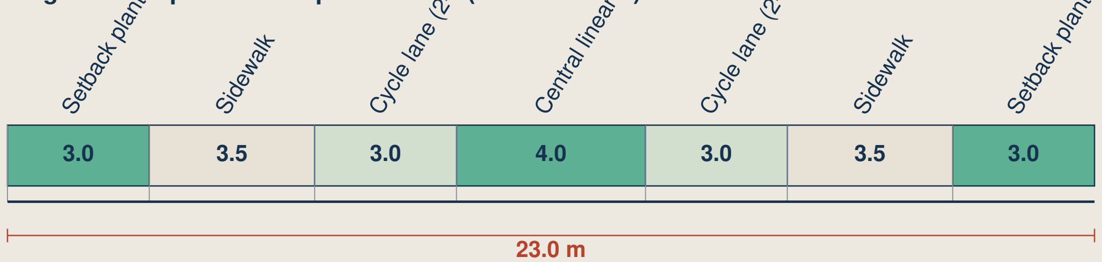
Key-area plan (diagrammatic, not a redline)

Recomputed: 104.32 ha | slow mobility 3,092 m | density 29.6 m/ha | public space 34.4%



Origin Works pedestrian spine section (YD-A to YD-C)



Closed to private vehicles; coloured permeable concrete cycle lanes; kiosk in the central garden.

key-areas-origin-works

Fig. 07 Origin Works, 104 ha

Recomputed: 104.32 ha | slow mobility 3,092 m | density 29.6 m/ha | public space 34.4%

- Blocks YD-A..YD-E: campus-adjacent incubation, release hall and demo day, open-source housing (youth rental), community service belt, open experiment ground.
- Pedestrian spine 1,100 m with a 4.0 m central linear garden; 23.0 m standard section.
- Public anchors: Examination Gate Plaza approx. 2,400 m², Origin Stele Garden approx. 1,200 m², RCK-02 kiosk.
- Closed to private vehicles; coloured permeable concrete cycle lanes throughout.

Each pocket park is approx. 600 m² and links three north-south pedestrian branches.

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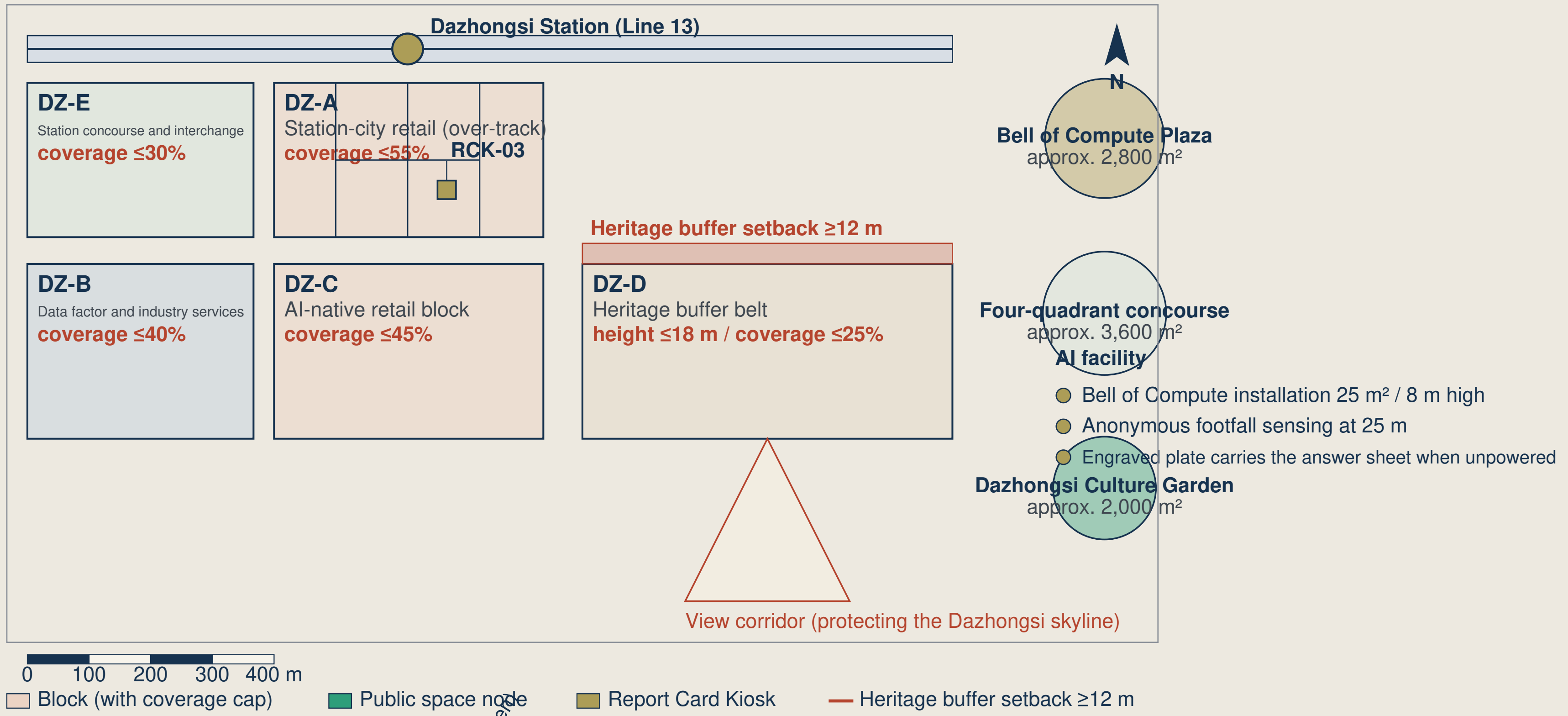
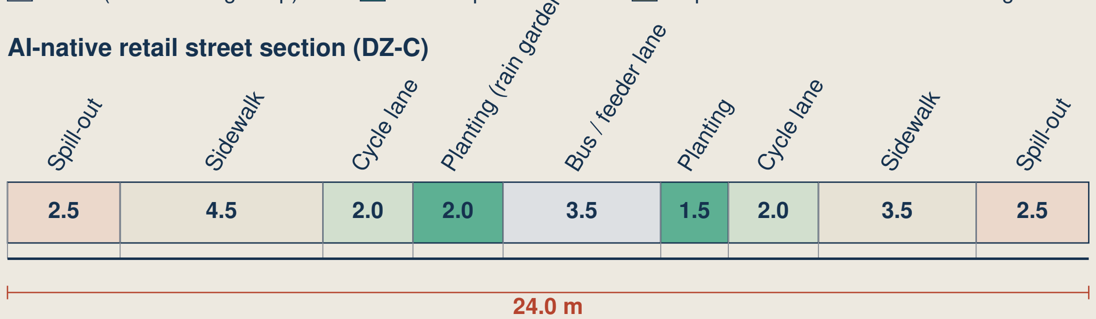
Key-area plan (diagrammatic, not a redline)

Recomputed: 72.05 ha | slow mobility 2,283 m | density 31.7 m/ha | green ratio 30.2%



AI-native retail street section (DZ-C)



Transit plus slow mobility, private vehicles excluded; demountable spill-out zones.

key-areas-dazhongsi

Fig. 08 Bell Hub, 72 ha

Recomputed: 72.05 ha | slow mobility 2,283 m | density 31.7 m/ha | green ratio 30.2%

- Blocks DZ-A..DZ-E: green compute, industry engine, AI-native retail, living, quadrant park.
- AI-native retail street with a 24.0 m section; heritage buffer setback ≥ 12 m from the Dazhongsi temple.
- Public anchors: Clock of Compute, the four-quadrant station concourse, RCK-03 kiosk.
- Transit plus slow mobility only; demountable spill-out zones are withdrawn at night to release pedestrian space.

The protection zone and construction control belt must be re-checked once official heritage data is published.

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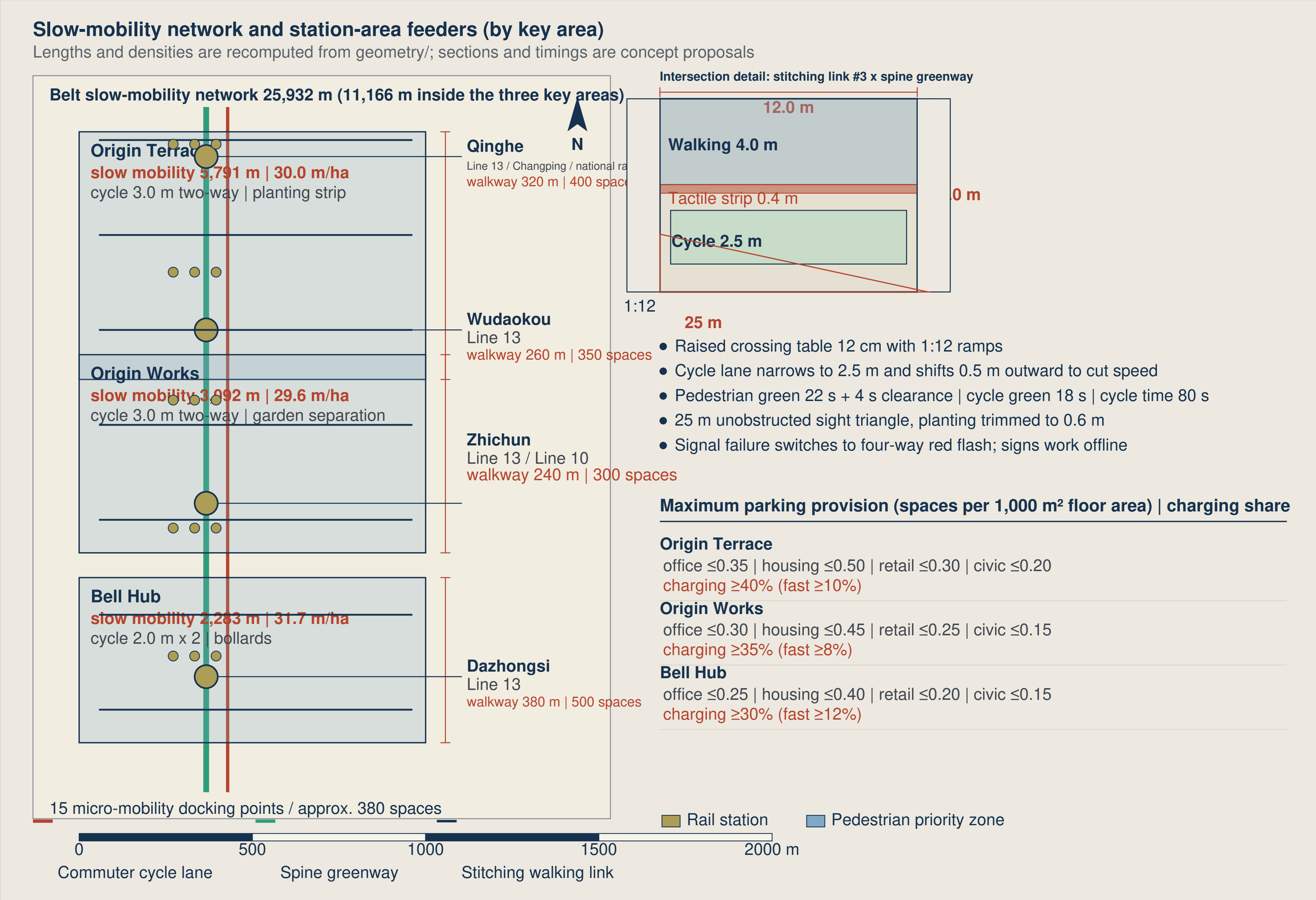

mobility-strategy

Fig. 09 Transport strategy

Belt slow-mobility network 25,932 m, of which 11,166 m lies inside the three key areas.

- Station feeders: Qinghe, Wudaokou, Zhichun and Dazhongsi, with covered walkways of 240–380 m.
- Intersection detail: raised crossing table 12 cm with 1:12 ramps and a 25 m unobstructed sight triangle.
- Signal failure switches to four-way red flash; signs keep working offline.
- Maximum parking provision per key area, with a fast-charging share of ≥ 30–40%.

Present dimensions and modal splits are concept proposals only, pending traffic counts, station OD data and rail interface data.

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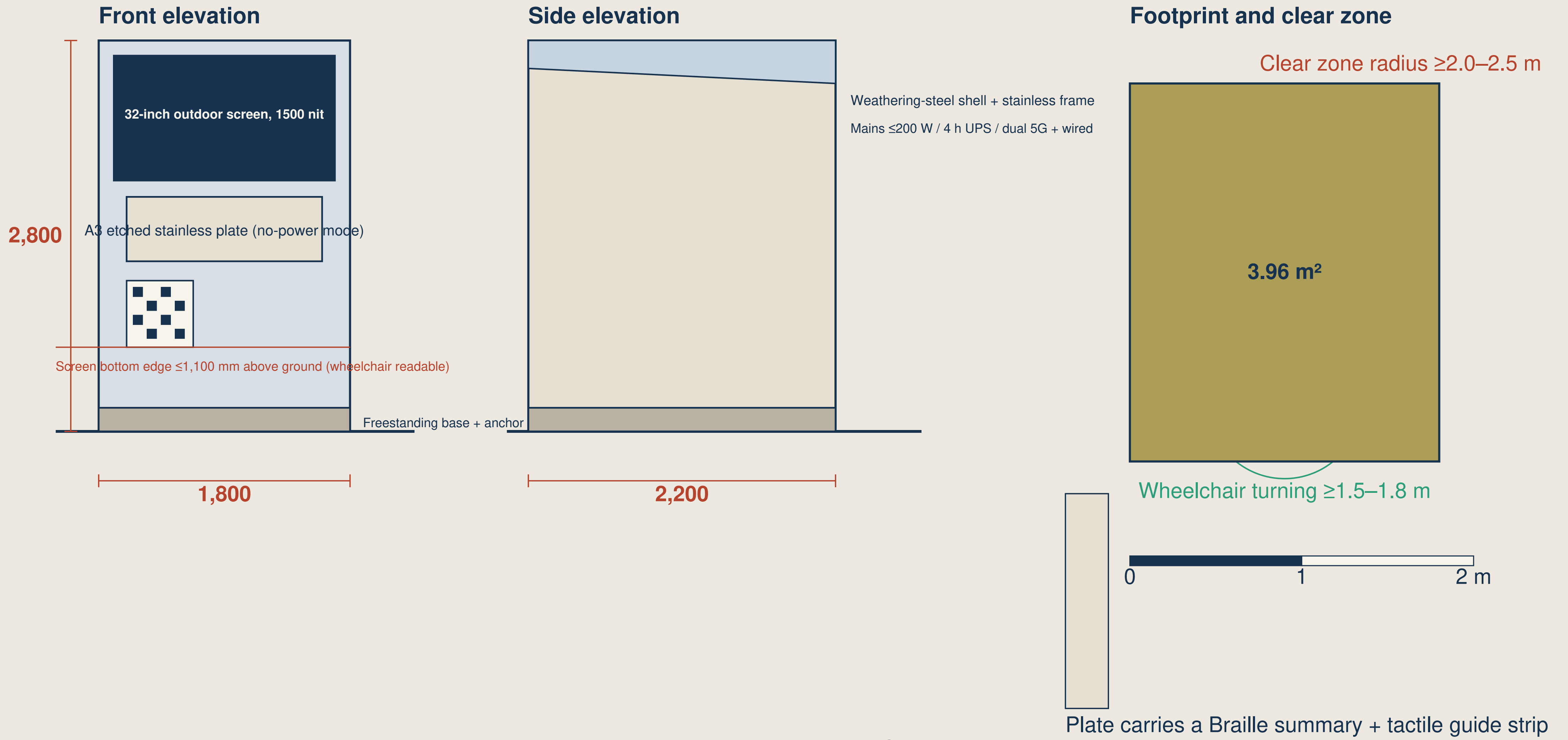
Report Card Kiosk: elevations and footprint (front / side / clear zone)

Overall 1,800 x 2,200 x 2,800 mm | footprint 3.96 m² | freestanding, demountable



Active mode shows the six answer-sheet items and human-takeover records; the plate and QR code stay readable when unpowered.

Deployment: RCK-01 Origin Terrace | RCK-02 Origin Works | RCK-03 Bell Hub | RCK-04 spine stitching link #3

report-card-kiosk

Fig. 10 Report Card Kiosk

1,800 × 2,200 × 2,800 mm | footprint 3.96 m² | freestanding and demountable

- Powered mode: 32-inch high-brightness outdoor display (1,500 nit) showing the F1–F6 status in real time.
- No-power mode: A3 etched stainless plate plus a physical QR code pointing to a static manifest.
- Accessibility: clear operating zone ≥ 2.0 m, wheelchair turning radius ≥ 1.5 m, screen centre 1,000–1,200 mm.
- Deployment: RCK-01 Origin Terrace | RCK-02 Origin Works | RCK-03 Bell Hub | RCK-04 spine stitching link #3.

The plate also carries a Braille summary and a tactile guide strip, so no smartphone is required.

1. Six talent personas

- Origination scientist: quiet, long-term exploration in dedicated labs with low electromagnetic noise.
- Open-source engineer: low-cost flexible co-working, with contributions recognised on the Origin Stele.
- AI startup founder: fast iteration and seamless access to test sandboxes and venture capital.
- Industry engineer: real-world industrial data and edge deployment environments.
- International talent: bilingual signage and seamless international life services.
- Local community resident: primary beneficiary of the green parks and of smart services that keep a human fallback.

2. Twelve scenario cards and four sandboxes

- 12 scenario cards covering intelligent traffic dispatch, adaptive park lighting, AI health kiosks, environmental microclimate sensing and smart street furniture.
- 4 industrial test sandboxes:
  - Embodied robotics sandbox (Origin Terrace)
  - Autonomous delivery and logistics corridor (Heritage Park)
  - Generative urban design sandbox (Origin Works)
  - Green compute and energy dispatch sandbox (Bell Hub)

Every card states the user task, the space, the responsibility mix, the minimum data, the non-AI path, the release evidence and the exit condition.

1. Three-phase implementation

- Phase I (2026–2028), origin initiation:
  - Greenway spine connection, Open Source Hall, Renzi Terrace and the first batch of 4 sandboxes.
- Phase II (2029–2031), smart mesh expansion:
  - All 7 stitch links, 4 smart-station TODs and full coverage of Report Card Kiosks.
- Phase III (2032–2035), global leadership:
  - A world-class self-reliant AI ecosystem and an international governance dialogue platform.

2. Standards and compliance matrix

- MOHURD Urban Design Management Measures
- Detailed Planning Drafting Guidelines
- Universal Accessibility Design Standard GB 55019-2021
- Interim Measures for Generative AI Services
- Elder-Friendly Smart Technology catalogue (background reference only, not a metric source)
- Strict privacy protocol: no raw biometric face data is transmitted to the cloud.

Tenure, permits, control planning, municipal utilities and heritage clearance are preconditions for any site start.